# CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): Rusting of iron is a chemical change. Reason (R): A new substance with different properties (iron oxide) is formed. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** Assertion (A): Dry HCl gas does not change the colour of dry blue litmus paper. Reason (R): HCl gas does not ionise in the absence of water to release $H^+$ ions. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**3.** Assertion (A): Sodium is stored in kerosene oil. Reason (R): Sodium reacts vigorously with oxygen and moisture of the air. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**4.** Assertion (A): Carbon compounds have low melting and boiling points. Reason (R): Intermolecular forces in covalent compounds are weak. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**5.** 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

**(a)** 44 g **(b)** 28 g **(c)** 12 g **(d)** 20 g

**6.** Equal volumes of HCl and NaOH solutions of the same concentration are mixed. The resulting solution has pH: 1

**(a)** 7 **(b)** 13 **(c)** 1 **(d)** 3

**7.** Metal X reacts with dilute HCl giving $H_2$ but does not react with cold or hot water. Metal Y reacts with cold water. Which is more reactive? 1

**(a)** cannot be said **(b)** X **(c)** both equally **(d)** Y

**8.** The balanced equation for the complete combustion of methane is: 1

**(a)** $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ **(b)** $CH_4 + 3\,O_2 \longrightarrow CO_2 + 2\,H_2O$ **(c)** $2\,CH_4 + O_2 \longrightarrow 2\,CO_2 + H_2O$ **(d)** $CH_4 + O_2 \longrightarrow CO_2 + H_2O$

**9.** A metal X turns black on heating in air. The black product, on passing hydrogen over it, turns reddish brown. X is: 1

**(a)** zinc **(b)** copper **(c)** magnesium **(d)** iron

**10.** A white powder X on adding water forms a hard solid mass. X is: 1

**(a)** plaster of Paris **(b)** bleaching powder **(c)** washing soda **(d)** baking soda

**11.** A student heats a green carbonate ore of copper. The black residue and gas are: 1

**(a)** Cu and $CO_2$ **(b)** $Cu_2O$ and $O_2$
**(c)** $CuO$ and $CO_2$ **(d)** $CuO$ and CO

**12.** A hydrocarbon X decolourises bromine water and has a molecular mass of 28 u. X is: 1

**(a)** ethyne **(b)** ethene **(c)** methane **(d)** ethane

**13.** **Assertion (A):** Burning of magnesium ribbon in air is a combination reaction. 1
**Reason (R):** In a combination reaction two or more substances combine to form a single product.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** Dry hydrogen chloride gas does not change the colour of dry blue litmus paper. 1
**Reason (R):** Acids produce $H^+$ ions only in the presence of water.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** Sodium is stored under kerosene oil. 1
**Reason (R):** Sodium reacts vigorously with oxygen and moisture of the air and may catch fire.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** 12 g of carbon burns completely in 32 g of oxygen. Find the mass of carbon dioxide formed and name the law applied. 2

**17.** When a few drops of conc. HCl are put on a piece of limestone, bubbles form. Name the gas, test and equation. 2

**18.** A metal X reacts with dilute HCl and also with NaOH producing hydrogen. Identify the nature of X's oxide with an example metal. 2

**19.** Write the balanced equation for the complete combustion of ethane and name the products. 2

**20.** Why is the mass of the residue less than the mass of lead nitrate taken when it is heated in an open tube? 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Balance $PbO + C \longrightarrow Pb + CO$ and identify the oxidising and reducing agents. 3

**22.** A student is given solutions of pH 1, 4, 7, 10 and 13. Which will turn phenolphthalein pink and which can neutralise a base? Arrange by acidity. 3

**23.** In the reaction $ZnO + C \longrightarrow Zn + CO$ identify the oxidising agent, reducing agent and the type of reaction. Why is it used for zinc? 3

**24.** Write the structural formula of propene and state two ways in which it differs from propane. 3

OR

Green coating forms on copper vessels kept in humid air. Name the coating, give its cause and one prevention. 3

**25.** A white solid X on heating gives $CO_2$ and a residue Y. Y dissolves in water, is basic and is used in whitewash. Identify X and Y and write equations. 3

OR

Why does iron rust faster near the sea? Explain the role of salt in rusting. 3

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## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** **Case study.** In a school laboratory, Rohan took about 2 g of green ferrous sulphate crystals in a dry boiling tube and heated it strongly. The crystals first lost water and turned white, then changed to a reddish-brown solid, and a smell like burning sulphur was noticed. 5

(i) What is the colour of the final solid residue?

(ii) Name the type of reaction taking place on strong heating.

(iii) Write the balanced chemical equation for the decomposition of anhydrous ferrous sulphate.

(iv) Name the two gaseous products and state whether the reaction is endothermic or exothermic, with a reason.

**27.** Four metals A, B, C and D: A displaces B, C and D from their salt solutions; B displaces C and D; C displaces D; D displaces none. Arrange them in order of reactivity and suggest a suitable extraction method for A and for D. Name one possible set of four metals. 5

OR

Explain conversion of ethanol to ethene, ethanoic acid and ethyl ethanoate with reagents and equations. Identify the oxidising agent and the catalyst. 5

**END OF QUESTION PAPER** **Total: 50 Marks**

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